

Unit 3.5 Implicit differentiation

Note Title

2012/04/13

Section 3.5 in Stewart p 209-217

There are two ways to define functions implicitly & explicitly.

Most of the equations we have been dealing with have been explicit.

such as $y = 2x - 3$

$$f(x) = y = \sin[x^2]$$

Consider The following

$$\sin 2y = \sqrt{x} y^2 + 2$$

$$\sqrt{y} = \cos(y^2 + 1) \sqrt{x}$$

implicit equations

Example 1 Find y' for $xy = 1$

$$\frac{d}{dx} [xy] = \frac{d}{dx} 1$$

$$\frac{d}{dx} [x] \cdot y + x \frac{dy}{dx} = 0$$

$$1 \cdot y + x \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{y}{x} = -\frac{1}{x^2}$$

Example Consider

$$x^2 + y^2 = 25$$

Explicit $y = \pm \sqrt{25 - x^2}$

Consider $y = -\sqrt{25 - x^2}$

$$\frac{dy}{dx} = -\frac{1}{2} (25 - x^2)^{-\frac{1}{2}} \cdot (-2x) = \frac{x}{\sqrt{25 - x^2}}$$

Implicit

$$x^2 + y^2 = 2S$$

$$\frac{d}{dx} [x^2 + y^2] = \frac{d}{dx} 2S$$

$$\frac{d}{dx} x^2 + \frac{d}{dx} y^2 = 0$$

$$2x + 2y \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{2x}{2y} = -\frac{x}{y} = \frac{-x}{-\sqrt{2S-x^2}}$$

Example Find $\frac{dy}{dx}$ if

$$\textcircled{1} 2x^3 + y^4 + x^5 y^3 = 1$$

$$\frac{d}{dx} [2x^3 + y^4 + x^5 y^3] = \frac{d}{dx} 1$$

$$6x^2 + 4y^3 \cdot \frac{dy}{dx} + 5x^4 y^3 + x^5 3y^2 \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} [4y^3 + x^5 3y^2] = -6x^2 - 5x^4 y^3$$

$$\frac{dy}{dx} = \frac{-6x^2 - 5x^4y^3}{4y^3 + x^5y^2}$$

$$\textcircled{2} \quad y \sin 2x = \cos y$$

$$\frac{dy}{dx} \sin 2x + y \cos 2x \cdot 2 = -\sin y \cdot \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{-y \cos 2x \cdot 2}{\sin 2x + \sin y}$$

$$\textcircled{3} \quad \sin(x+y) = y^2 \cos x$$

$$\cos(x+y)[1+y'] = 2y \cdot y' \cos x - y^2 \sin x$$

$$\cos(x+y) + y' \cos(x+y) - 2y \cdot y' \cos x = -y^2 \sin x$$

$$y' = \frac{-y^2 \sin x - \cos(x+y)}{\cos(x+y) - 2y \cos x}$$

Derivatives of inverse trigonometric functions

$$\bullet \quad \frac{d}{dx} [\arcsin x] = \frac{1}{\sqrt{1-x^2}}$$

Proof:

$$y = \arcsin x \Leftrightarrow \sin y = x \quad \text{for } -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$$

$$\frac{d}{dx} [\sin y] = \frac{d}{dx} x$$

$$\cos y \cdot \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{\cos y}$$

$$= \frac{1}{\sqrt{1 - \sin^2 y}} = \frac{1}{\sqrt{1 - x^2}}$$

$$\begin{aligned} \cos^2 x + \sin^2 x &= 1 \\ \cos x &= \sqrt{1 - \sin^2 x} \end{aligned}$$

$$\bullet \frac{d}{dx} [\tan^{-1} x] = \frac{1}{1+x^2}$$

Proof:

$$y = \arctan x \Rightarrow \tan y = x \text{ and } y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

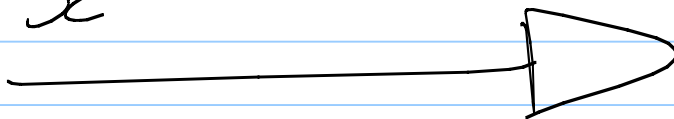
$$\frac{d}{dx} \tan y = \frac{d}{dx} x$$

$$\sec^2 y \cdot \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{\sec^2 y}$$

$$= \frac{1}{1 + \tan^2 y}$$

$$= \frac{1}{1 + x^2}$$



Examples Find derivatives:

$$1) f(x) = \sin(x^2)$$

$$f'(x) = \cos(x^2) \cdot 2x$$

$$2) \quad f(x) = \arcsin(x^2)$$

$$f'(x) = \frac{1}{\sqrt{1 - (x^2)^2}} \cdot 2x$$

$$3] \quad y = \tan e^{3x}$$

$$y' = \sec^2(e^{3x}) \cdot e^{3x} \cdot 3$$

$$4] \quad y = \arctan(e^{3x})$$

$$y' = \frac{1}{1 + (e^{3x})^2} \cdot e^{3x} \cdot 3$$

$$5) y = \arctan(\sqrt{t+1})$$

$$\frac{dy}{dt} = \frac{1}{1 + (\sqrt{t+1})^2} \cdot \frac{1}{2} (t+1)^{-\frac{1}{2}}$$

$$6) y = \arcsin\left(\frac{1}{t}\right)$$

$$\frac{dy}{dt} = \frac{1}{\sqrt{1 - \left(\frac{1}{t}\right)^2}} \cdot -t^{-2}$$

Example

Find the equation of the tangent line to the curve

$$x \sin y - y \cos 2x = 2x \quad \text{at the point} \quad \left(\frac{\pi}{2}, \pi\right)$$

$$\begin{aligned} & 1 \cdot \sin y + x \cos y \cdot \frac{dy}{dx} - \frac{dy}{dx} \cos 2x \\ & + y \sin 2x \cdot 2 = 2 \end{aligned}$$

$$\frac{dy}{dx} = \frac{2 - \sin y - y \sin 2x \cdot 2}{x \cos y - \cos 2x}$$

$$\left. \frac{dy}{dx} \right|_{\substack{x=\pi \\ y=\frac{\pi}{2}}} = \frac{2 - \sin\left(\frac{\pi}{2}\right) - \frac{\pi}{2} \sin 2\left(\pi\right) \cdot 2}{\pi \cos \pi/2 - \cos\left(2 - \frac{\pi}{2}\right)}$$

$$\left(y - \frac{\pi}{2}\right) \approx f'(\pi) (x - \pi)$$

Example

Find The values of $x \in [0, \pi]$ where The tangent line to $f(x) = \sin 2x - \cos 2x$ is horizontal

Example

Find the equation of the tangent line
to $f(x) = x\sqrt{x^2+1}$ at the point
 $x=1$.

